

New issue



[Performance]: Improve Prefix Cache Hit Rate and Reduce Dirty Cache Impact #24394

Closed as not planned

Labels

performance

stale



CLFutureX opened on Sep 7, 2025

Contributor



Proposal to improve performance

Optimization Proposal: Improve Prefix Cache Hit Rate and Reduce Dirty Cache Impact

Current Background

Currently, vLLM uses prefix caching for matching. The earlier a request is processed, the higher its hit rate.

Therefore, when evicting and releasing blocks, they are reversed before being added to the free list, implementing eviction from the tail to improve block reuse rate.

However, between multiple adjacent requests, the overall block distribution is not reversed, preventing adjacent multiple requests from utilizing this characteristic.

If reverse release between multiple requests can be achieved, it will undoubtedly improve the block hit rate.

Difficult-to-Solve Dirty Cache Problem

The current design, when encountering the release of an ultra-long request, causes the cached blocks of multiple adjacent requests ahead to be prioritized for eviction, which is a typical dirty cache problem.

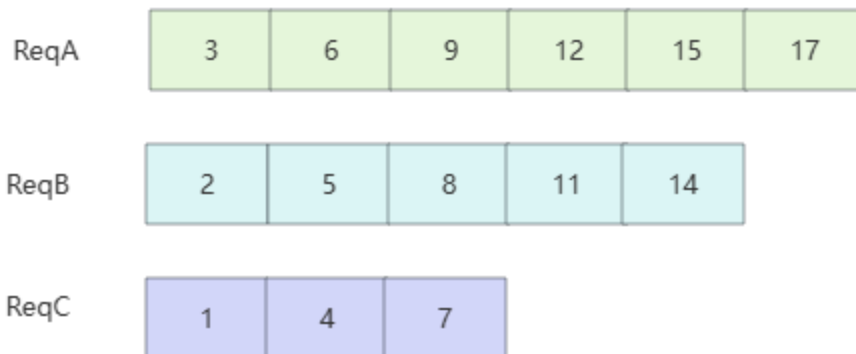
Implementing reverse release between multiple adjacent requests could significantly alleviate the dirty cache problem.

Solution

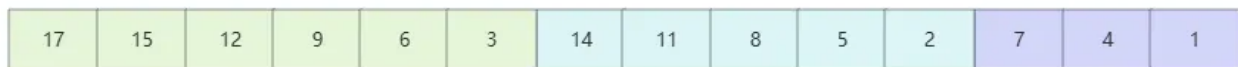
How can we leverage this characteristic during the release of multiple requests to improve overall cache hit rate and mitigate the dirty cache phenomenon?

Delayed Batch Release: During release, first temporarily store them. When a specified quantity is reached or when available blocks are insufficient, then uniformly release these requests, thereby achieving reverse release between multiple adjacent requests.

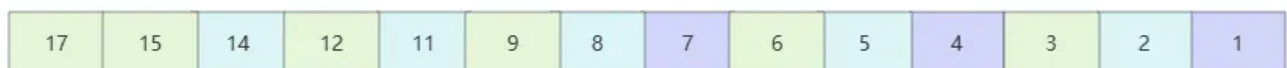
example:



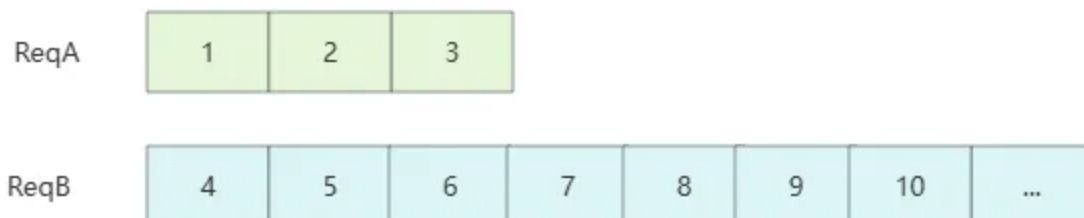
optimize before:



optimize after:



dirty cache problem:



optimize before:

...	10	9	8	7	6	5	4
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optimize after:

...	7	3	6	2	5	1	4
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Report of performance regression

No response

Misc discussion on performance

No response

Your current environment (if you think it is necessary)

The output of ``python collect_env.py``



Before submitting a new issue...

- ☒ Make sure you already searched for relevant issues, and asked the chatbot living at the bottom right corner of the [documentation page](#), which can answer lots of frequently asked questions.



CLFutureX added **performance** on Sep 7, 2025



CLFutureX mentioned this on Sep 7, 2025

[\[Performance\]: Improve Prefix Cache Hit Rate and Reduce Dirty Cache Impact #24395](#)



github-actions bot on Dec 8, 2025 – with [GitHub Actions](#)



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 **github-actions** added **stale** on Dec 8, 2025

 github-actions **bot** on Jan 10 – with [GitHub Actions](#) ...

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 **github-actions** closed this as not planned on Jan 10

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Metadata

Assignees

No one assigned

Labels

performance **stale**

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or pull requests

Participants

